

Reviewer Pack — Minimal Order of Arithmetic Hierarchy invariants and Resolut...

Entry 373395adc224 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-06-05 02:04:32 UTC

Minimal Order of Arithmetic Hierarchy invariants and Resolution Proof Length Correlation

Entry ID: 373395adc224 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-05 02:04:32 UTC

1. Conjecture

Field A (mathematical branch): Arithmetic Hierarchy Theory **Field B** (complexity object): Resolution Proofs (Proof Complexity)

Statement:

For every Tseitin formula φ_G with n variables, the resolution proof length of φ_G is linearly correlated with the order of the arithmetic hierarchy invariants associated with its clauses, such that the maximum order among these invariants is $\Theta(n^2 \log n)$.

Rationale (proposer's reasoning):

Arithmetic Hierarchy Theory provides a framework for classifying functions by their complexity, which may reveal hidden structures in proof complexity. If this correlation holds, it suggests a deep connection between arithmetic hierarchy and proof length, potentially leading to new lower bounds or insights into the resolution proof system.

Taxonomy category: Arithmetic_Hierarchy_Theoretic (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): f1e0a8b8be7b6e1d

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if the correlation coefficient between the maximum order of the arithmetic hierarchy invariants and the resolution proof length is ≥ 0.9 for at least 80% of the generated Tseitin formulas with $n \leq 40$ variables, and all calculations are consistent within a standard deviation of $\leq 5\%$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	1.00	SAFE	UNCERTAIN
KARP_LIPTON	SAFE	1.00	SAFE	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - intitle:Arithmetic Hierarchy Theory AND resolution proof length - resolution proofs AND complexity theory AND 'arithmetic hierarchy' - invariant order AND Tseitin formula AND proof complexity

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=0, elapsed=0.0s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def tseitin_formula(n):
        clauses = []
        for i in range(1, n + 1):
            clauses.append([i])
            for j in range(i + 1, n + 1):
                clauses.append([-i, -j, i + j])
        return clauses

    def arithmetic_hierarchy_order(clause):
        # Placeholder function to compute the order of an arithmetic hierarchy invariant
        # This is a dummy implementation and should be replaced with actual logic
        return len(clause)

    def resolution_proof_length(clauses):
        # Placeholder function to compute the length of a resolution proof
        # This is a dummy implementation and should be replaced with actual logic
        return len(clauses) * 2

    n = random.randint(5, 40)
    clauses = tseitin_formula(n)
    max_order = max(arithmetic_hierarchy_order(clause) for clause in clauses)
    proof_length = resolution_proof_length(clauses)

    return {
        "metric_name": "correlation_coefficient",
        "metric_value": max_order / proof_length,
        "instances_tested": 1,
```


9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge_falsified | original: The correlation coefficient between the maximum order of the arithmetic hierarchy invariants and the resolution proof length did not meet the threshold | next: Revise the test code to compute actual values for 'arithmetic_hierarchy_order' and 'resolution-proof_length'. Then, retest the conjecture using the revised code to determine if it holds or not.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	19500
2	preregistration	ollama_remote	glm4:latest	0	0	9495
3	novelty	ollama_remote	glm4:latest	0	0	8210
4	novelty	ollama_remote	glm4:latest	0	0	9460
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11361
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9187
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11745
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7270
9	critic	ollama_remote	glm4:latest	0	0	21867
10	judge	ollama_remote	glm4:latest	0	0	11348

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 119444 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in `research/audit/373395adc224.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/373395adc224.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/373395adc224.tar.gz` (if generated)